

Project 2

STAT 207 - Data Science Exploration

Due: Tuesday, October 21 by 11:59 pm on GitHub

Main Goal of Analysis

The main goal of this project is to use a dataset to make inferences about a population, quantifying uncertainty in the process. The secondary goal is to contrast the inferences with summaries for a sample.

You are required to perform two main analytical tasks:

1. Estimating a parameter with a confidence interval
2. Defining and testing theories about an unknown parameter.

Additional descriptions for these tasks can be found later in this document. If you think that there are additional questions or analyses that would add additional insights to your overall research goal, you're more than welcome to pursue these in addition to what is required in this document.

Project Format

Your project will be submitted as a written report to GitHub. This report is worth 35 points. Additional details for the report can be found below.

Group Structure

You must work in a group of either 2 or 3 students, and your group should all be enrolled in the same lab section.

- If you work with a group of 3, you must do at least 25% of the work in order to get full credit.
- If you work with a group of 2, you must do at least 33% of the work in order to get full credit.

Dataset Options

You can choose your own dataset, or you can use the supplied dataset below. There are several places you can go to find interesting datasets, but here are some places to start:

- <https://www.kaggle.com/datasets>
- <https://corgis-edu.github.io/corgis/csv>
- <https://archive.ics.uci.edu/ml/datasets.php>
- <https://github.com/fivethirtyeight/data>
- For sports data, you may choose to explore (nflfastr.com for NFL, billpetti.github.io/baseballr for MLB, cfbfastr.sportsdataverse.org/index.html for CFB, and sportsdataverse.org for more sports data).

Choosing your own data:

If you choose your own data, it must meet the following specifications.

1. It must be smaller than 25 MB (25 megabytes, or 25,000 kilobytes). If your file is measured in kilobytes, it should be small enough for the purposes of our projects. If your file is measured in GB, it is too large and you will need to reduce your data for the project. This file size restriction is to ensure that you can push your file to GitHub. You can stop by office hours if you find a larger dataset, and we can help you take a random sample of the data to fulfill this data size requirement.
1. It must have at least two variables total (will need at least five for Project 3)
 - a. Variables that have uninformative information don't count and won't be useful. Examples of uninformative variables include those that provide the unit of observation (a row name or row id) or are a linear combination (sum, product) of other variables in the datasets. If you aren't sure, come ask!
2. It must have at least one categorical variable
 - a. For this categorical variable, you will use only two levels; that is, you will adjust it to a logical variable. You may already have a logical variable in your data, or you may need to explicitly create a logical variable. Options to do so include:
 - i. Filter your data to include only two levels of a categorical variable
 - ii. Explore one level vs. all other levels of a categorical variable
 - iii. Combine similar levels of a categorical variable, resulting in two levels. For example, you could make "year in school" into "upperclass?".
3. It must have at least one quantitative variable
4. It must have at least 50 rows

You may continue using the same dataset from Project 1, or you could change to a different dataset.

Provided dataset:

1. Video Games Data (video_games.csv)
 - a. This dataset has information about the sales and playtime of over a thousand video games released between 2004 and 2010. The playtime information was collected from crowd-sourced data on "How Long to Beat"
 - b. This was originally collected and curated by Dr. Joe Cox
 - c. This data was originally downloaded on 2/8/2024 from here:
<https://researchportal.port.ac.uk/en/datasets/video-games-dataset>
 - d. Read more about this data here:
<https://researchportal.port.ac.uk/en/publications/what-makes-a-blockbuster-video-game-an-empirical-analysis-of-us-s>
 - e. Note: while reading in this data to Python, you will need to use the argument `encoding = 'unicode-escape'` with code like `df = pd.read_csv('video_games.csv', encoding = 'unicode-escape')`

Project Report Specifications

Deadline: Tuesday, October 21 by 11:59 pm on GitHub

Format:

- Jupyter notebook
- This should be a clean data analysis report that you could submit to an employer or client (not a homework assignment). At the very least, your report should have a title, headings for each section, and be written in paragraphs and with complete sentences.
- You can use and modify the attached `project_02_template.ipynb` file as a template for this report if you'd like. You can add and delete as many cells as you'd like in this file.

1. Introduction [4 points]

Goal: In your introduction, you should orient the reader to what they are about to read. This will help to prepare the reader, so that they can figure out how to connect the different components that they will read.

While working towards this goal, you should complete/address the following:

- a. Title: Give your research report a title
- b. Dataset introduction: In a couple of sentences, you should briefly introduce the reader to the context of your dataset and the available data.
- c. Populations and Samples: Does your data represent a population or a sample? What is the corresponding population of interest? If your data represents a population, you should take a random sample of your data for this project.
- d. Research Questions: You will answer two **sets** of research questions. In your introduction, you should state your research questions. These should be the same research questions stated at the beginning of section 2 and at the beginning of section 3.
- e. Contextual Importance: You should describe why you (or someone else) would be interested in the answer to these research questions. For example, how could the answer be used? You can use creativity and imagination when describing a situation where this answer would be helpful.

2. Confidence Interval Analytical Task [10 points]

Goal: For your confidence interval analytical task, you will use your logical variable. You will explore the behavior of this variable *in the dataset* (descriptive analytics) and *in the population* (inference).

While working towards this goal, you should complete/address the following:

- a. State your research question: Your research question should have two parts. For example, you might ask "What is the proportion for the logical variable in this dataset? What are a range of reasonable values for the proportion for this variable in the underlying population?"
 - i. Remember, descriptive analytics only involves describing the dataset that you have, so your first research question should be *just* about the data. Then, inferential statistics involves answering research questions about populations given a random sample from that population. Your second

research question should reference the population from which your data were collected.

- b. Dataset cleaning: Be sure to clean the data for your variable of interest. You may need to adjust how your variable is recorded or filter your data to be sure it only contains relevant data for your population of interest. Be sure to describe these steps, along with any limitations.
- c. Descriptive analytics:
 - i. Report your sample size.
 - ii. Numerical summaries: Calculate at least one appropriate numerical summary for your variable of interest.
 - iii. Interpret your numerical summaries. That is, report your results in a sentence, including context.
- d. Create a confidence interval:
 - i. Select an appropriate confidence level
 - ii. Simulate a sampling distribution for your proportion
 - iii. Use this simulated sampling distribution to estimate your confidence interval
- e. Interpret your confidence interval.
 - i. Provide a formal interpretation for the confidence interval.

3. Hypothesis Testing Analytical Task [15 points]

Goal: For your hypothesis testing analytical task, you will use your quantitative variable. You will explore the distribution of this variable *in the dataset/sample* (descriptive analytics) and *in the population* (inference).

While working towards this goal, you should complete/address the following:

- a. State your research question: For instance, you could ask “What are summary measures of my variable in this dataset? Which of two competing theories about my population is supported by the data in the sample?”
- b. Dataset cleaning: Be sure to clean the data for your variable of interest. Describe these steps, along with any limitations.
- c. Descriptive analytics:
 - i. Numerical summaries: Calculate appropriate numerical summaries to observe features about your quantitative variable of interest.
 - ii. Visualization: Generate an appropriate visualization to observe the distribution of your variable of interest.
 - iii. Interpret the results of these two descriptive analytic tasks.
 - iv. What is the most appropriate measure of center to describe your variable of interest?
- d. Perform a hypothesis test:
 - i. State your hypotheses, including a definition and description of your parameter of interest. You may select the hypotheses you wish to test.
 - ii. Select a significance level.
 - iii. Check the conditions for this test.
 - iv. Simulate a sampling distribution for your statistic of interest.

- v. Calculate a p-value for this test based on the sampling distribution.
- vi. Evaluate your results to make a decision and state a conclusion.
- e. Interpret your p-value:
 - i. Provide a formal interpretation for the p-value.

4. Conclusion [4 points]

- a. Summarization: Summarize your confidence interval and hypothesis test tasks in the conclusion. Provide about a paragraph. (This will likely be a restatement of what you have already included in your report).
- b. Limitations: What limitations did you face in your analysis, results, or interpretations? What challenges did you face in your data analysis? What contextual information is important before you make strong claims from these results? How might these affect how the person you described in the introduction uses these results?
- c. Future work: If you (or someone else) were to conduct future work based on these analyses, what kind of research questions or analyses might that entail?

The remaining **2 points** of this project will be graded on writing quality, clarity, conciseness, and professional and neat formatting of the report.

Intended Audience/Reader of your Project

The intended audience of your report/presentations should be someone who has the same level statistical/python knowledge as you and your STAT 207 classmates. **Theoretically, you should be able to send your report to one of your classmates (who is not on your team), and they should be able to understand everything that you did and the claims that you are making.**

Grading

In addition to being graded for correctness and completion (as noted), this project will be graded on a qualitative basis. Qualitatively, we will be looking for the following things:

- **Clarity about Analyses, Algorithms, and Data Choices**
 - Someone who has taken a STAT207-level class should be able to read through your report and easily be able to do the following:
 - Replicate what you did in your analyses.
 - Know why you made the choices that you did in your analyses.
- **Clarity about Motivation (i.e. the “so what?”) of your analyses**
 - Beginning of the Report
 - Someone who is **about to** read the body of your report should be able to clearly answer the questions:
 - Why should I (or someone else) care about the report that I am about to read?
 - What research questions do they intend to answer?
 - How do these research questions relate to their motivation?
 - Therefore, in the introduction of your report you should make this clear.
 - Middle of the Report:
 - While **in the middle of** your report, your audience should be able to clearly answer the question:
 - How do each of these analyses/algorithms/data choices that they’re making/using tie back into the overarching motivation of this whole analysis?
 - Therefore, each new analysis/model/algorithm/data choice that you make, you should explain this and make it clear to your audience.
 - End of the Report:
 - Someone who has **just finished** reading your report should be able to clearly answer the questions:
 - Why should I (or someone else) care about the analysis that I just read?
 - Did their analyses and conclusions answer the research questions that they stated at the beginning of the report? If so, how? What were the answers to these research questions?
 - How would the results/answers to these research questions be useful to someone?
 - Therefore, in the conclusion of your report you should make this clear.